22 A loudspeaker is playing music in a room. The door to the room is open and has a width of 0.80 m.

Sound waves of many different frequencies pass through the doorway and diffract. The speed of sound in air is $340\,m\,s^{-1}$.

Which frequency of sound wave diffracts the most as it passes through the doorway?

A $2.4 \times 10^{-3}\,Hz$

B $8.0 \times 10^{-1}\,Hz$

C $2.7 \times 10^{2}\,Hz$

D $4.3 \times 10^{2}\,Hz$